UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	:	CSC429
EXAMINATION	:	NOVEMBER 2026
TIME	:	

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC429 mock paper (November 2026). Generated answers show working. Real past-paper questions are reproduced with their source reference; no answer is invented where the original scheme is unavailable.

2. Generated marking points state the expected answer and the allocation of marks. Real-bank questions retain their source paper and page reference.

3. Part A = 4 x 5 + 6 x 10; Part B = 1 x 20; total = 100 marks.

4. Generated slots: 2. Real-bank slots: 9. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 20 unique CSC429 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

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PART A — MARKING GUIDE

QUESTION A1 (5 marks)

Topic: Computer generations and history

Question 1 is based on the following Figure 1.
The development of component X led to a drastic increase in speed and efficiency of computers. Component X is a single, complete electronic semiconductor circuits contained on pieces of silicon, and sometimes called as chips.
Figure 1: Component X
a) What is the name of component X?
b) In which computer generation was component X introduced?
c) Who is the inventor of component X?
d) State ONE (1) input device and ONE (1) output device that was introduced in the same generation as component X.

Source: CSC429 (12).pdf, page(s) 2

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A2 (5 marks)

Topic: Instruction formats and addressing modes

Given the following Figure 2.
Instruction
Opcode
Address A
Memory
Operand
Figure 2
Label the addressing technique illustrated in Figure 2.
a)
Explain how the addressing technique in 9(a) is used.
b)
List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A3 (5 marks)

Topic: Data representation and number systems

Convert 2.5 to IEEE 754 single-precision format. Give the answer in hexadecimal and show the sign, exponent and mantissa fields.

Answer: 2.510 = 0x40200000 (sign=0, exponent=10000000, fraction=01000000000000000000000)

- *1 mark for the sign bit.*
- *1 mark for the biased exponent.*
- *1 mark for the 23-bit fraction.*
- *1 mark for the correct 32-bit grouping.*
- *1 mark for the final hexadecimal value.*

QUESTION A4 (5 marks)

Topic: Parallel and multiprocessor architecture

a) How classification of parallel processor is constructed?

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b) Give ONE (1) example for each parallel processor architecture.

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A5 (10 marks)

Topic: Data representation and number systems

Answer both parts for the same expression. Assume an 8-bit byte.
Expression: 28 - (-78)

a) Calculate 28 - (-78) using sign and magnitude.

Answer: left 28 -> 0 0011100; right -78 -> 1 1001110; 28 - (-78) = 106 -> sign-magnitude 0 1101010

- *1 mark for operand representations.*
- *1 mark for sign handling.*
- *1 mark for magnitude arithmetic.*
- *1 mark for the result sign bit.*
- *1 mark for the final byte.*

b) Calculate 28 - (-78) using 1's and 2's complement.

Answer: left 28 -> 1's 00011100, 2's 00011100; right -78 -> 1's 10110001, 2's 10110010; 28 - (-78) = 106 -> 1's 01101010, 2's 01101010

- *1 mark for the positive operand in binary.*
- *1 mark for the 1's complement of the negative operand.*
- *1 mark for adding 1 to form the 2's complement.*
- *1 mark for the correct 8-bit result.*
- *1 mark for converting the result back to decimal.*

QUESTION A6 (10 marks)

Topic: Memory hierarchy and cache

Replacement algorithm is used to determine which entry in a cache to flush to main memory when it is desired to cache a new block of data.
a) Which mapping function does not need to have a replacement algorithm?
b) Describe FOUR (4) types of replacement algorithm.

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A7 (10 marks)

Topic: Computer architecture and CPU organisation

Computer consists of four main components as illustrated in Figure 1. Each component has its own function in ensuring that the computer system works well. Question 4 is based on Figure 1.

Figure 1 : Components of computer

a) Explain THREE (3) types of system interconnection.
b) State TWO (2) sub-components of central processing unit and its function.

Source: CSC429 (3).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A8 (10 marks)

Topic: Computer architecture and CPU organisation

Processor contains a number of registers used for temporary storage of data. Describe FIVE (5) types of registers.

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A9 (10 marks)

Topic: Input/output and interrupts

a)
Define interrupt.
b)
Differentiate between interrupt and direct manipulation access.
c)
Given the following scenario.
Dilan is surfing woolworth.com to buy groceries online. While he is comparing prices of products in woolworth.com and one other website, two messages popped up onto his screen. Figure 2 (a) and (b) illustrate the conveyed messages.

Are you sure you want to
cancel this request?

There is a hardware event failure
in this computer. Please reboot
this computer now.

(a)
(b)
Figure 2: The pop-up messages
i)
Identify the type of interrupt in Figure 2(a) and Figure 2(b) with proper justification.
ii)
While interruptroutinefor Interrupt Ais executing, it is interrupted by interrupt Bwhich has a higher priority than interrupt A. What will happen to interrupt routine A?

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A10 (10 marks)

Topic: Input/output and interrupts

a) When is an I/O module installed?
b) What will happen if the I/O module is not installed?
c) Explain ONE (1) category of I/O module
d) What does it mean by 'hot connection'?
e) List THREE (3) common interfaces to external devices.

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

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PART B — MARKING GUIDE

QUESTION B1 (20 marks)

Topic: Assembly-language programming

Question 1 is based on Figure 1 and Table 1.
z=A2-B*C+jl_
D+5
Figure 1
write an appropriate program using correct assembly language notations to compute the equation snbwn in Figure 1 using instructions for zero-address, one-address, and three-address. The available instructioni are provided in Tabte 1. The register labeled A, B, C, and D are fixed, white registers x, y, and z can be used temporarily. The final answer must be stored in register z.
Table 1: Machine instruction
Three-address
One-address
Zero-address
ADD (p,q,r)
LOAD p
PUSH p
suB (p,q,r)
STOR p
POP p
MUL (p,q,r)
ADD p
ADD
DIV (p,q,r)
SUB p
SUB
MUL p
MUL
DIV p
DIV

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END OF MARKING SCHEME

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